THE SELECTION RULE AND GROUP REPRESENTATION METHOD IN ELECTRON-PHONON INTERACTION

HAIXUAN LIN

Abstract. It is well known that the group representation method can be used to determine whether a given transition is allowed. However, for electron-phonon interactions in a crystal — an infinite system — the symmetry group is not finite, and traditional representation theory is not a sufficiently powerful tool. In this work, we develop a rigorous and general method from the ground up to address this problem.

1. Introduction

The problem begins with the scattering amplitude for electron-phonon interaction:

$$g_{fi}^{\nu}(\mathbf{k},\mathbf{q}) := \langle \psi_{f\mathbf{k}+\mathbf{q}} | \hat{V}_{\nu\mathbf{q}} | \psi_{i\mathbf{k}} \rangle , \tag{1}$$

where $\mathbf{k}$ is the initial electron momentum, $\mathbf{q}$ is the phonon momentum, and i, f, and ν denote the initial and final electron band indices and the phonon mode, respectively. $|\psi_{i\mathbf{k}}\rangle$ and $|\psi_{f\mathbf{k}+\mathbf{q}}\rangle$ are the initial and final states of the electron before and after the phonon-induced Kohn–Sham perturbation potential. Our task is to determin whether (1) is allowed by symmetry or not by using group representation method.

2. Precise definition of Our Problem

For further work, we need to give a precise definition of our problem, including to clarify what is "allowed by symmetry", and to find out concrete groups and their representation space and so on.

By saying symmetry, we are discussing "something left invariant after doing some operations". The group is the collection of those operations, so we are supposed to find out those group left (1) invariant.

Let $\hat{H}$ be the single-body electron Hamiltonian, $\mathcal{H}$ the electron states Hilbert space and $SG := \left\{ \left(\hat{R} \middle| \mathbf{w} \right) \right\}$ the space group with action on Hilbert space defined intrinsically by $(g \cdot \psi)(\mathbf{r}) := \psi\left(g^{-1} \cdot \mathbf{r}\right)$ or more clearly by Dirac notation:

$$(g \cdot \psi)(\mathbf{r}) = \langle \mathbf{r} | g \cdot |\psi\rangle = (\langle \mathbf{r}| \cdot g) |\psi\rangle = \left(g^{-1} \cdot |\mathbf{r}\rangle\right)^{\dagger} |\psi\rangle \tag{2}$$

Date: Aug 7, 2026.

with $g \cdot |\mathbf{r}\rangle = |g \cdot \mathbf{r}\rangle$, giving definition of expression $g \cdot |\psi\rangle$ by defined on general complete basis $\{|\mathbf{r}\rangle\}$, then according to Bloch's theorem, there is a direct sum decomposition:

$$\mathcal{H} = \bigoplus_{\mathbf{k}} \mathcal{H}_{\mathbf{k}} \tag{3}$$

where

$$\mathcal{H}_{\mathbf{k}} \coloneqq \left\{ |\psi_{n\mathbf{k}}\rangle : \hat{T}_{\mathbf{w}} |\psi_{n\mathbf{k}}\rangle = \mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{w}} |\psi_{n\mathbf{k}}\rangle \right\}, \tag{4}$$

and the n presents any other quantum numbers, $\hat{T}_{\mathbf{w}}$ is the translation operator of vector $\mathbf{w}$. Due to the electron states in (1) are label by momentum as good quatum numbers, the proper representation space are some $\mathcal{H}_{\mathbf{k}}$'s and thus we need to find out the stabilizer group $G_{\mathbf{k}} \coloneqq \mathrm{Stab}_{SG}(\mathcal{H}_{\mathbf{k}})$ (called little group) that leave $\mathcal{H}_{\mathbf{k}}$ be $G_{\mathbf{k}}$-invariant.

The primitive definition of SG naturally involves the group action on real space, but now we need to work under reciprocal space so it urges us to give a natural group action of G on momentum. Let $L \coloneqq \{(1|\mathbf{r})\} \trianglelefteq SG$ be the latiice system in real space and $\widehat{L} \coloneqq \mathrm{Hom}(L, U(1))$ the Pontryagin dual with modulo 2π equivalence — physically speaking, the reciprocal space with modulo reciprocal lattice vector, and if you will, any element $\chi \in \widehat{L}$ can be express use Dirac notation of momentum by $|\chi\rangle \propto |\mathbf{k}\rangle$ — it is a 3 dimensional torus due to Born–von Karman boundary condition. Define conjugate action of SG on L:

$$\left(\hat{R}\middle|\mathbf{w}\right) \cdot (1|\mathbf{r}) \coloneqq \left(\hat{R}\middle|\mathbf{w}\right)(1|\mathbf{r})\left(\hat{R}\middle|\mathbf{w}\right)^{-1} = \left(1\middle|\hat{R}\mathbf{r}\right), \tag{5}$$

and we notice $\mathbf{w}$ vanishes, which means this conjugate action is trivial on L itself and thus go through quotient group $PG \coloneqq G/L$ — the so-called point group. We thus can define the action of SG on L by defining so of PG on L: $\forall g = \left(\hat{R}\middle|\mathbf{w}\right) \in SG$, or you can consider its quotient image $\hat{R} \in PG$, and $\forall \chi \in \widehat{L}$, we imitate (2) to define

$$(g \cdot \chi)(\mathbf{r}) \coloneqq \chi\left(g^{-1}\mathbf{r}g\right) = \chi\left(g^{-1} \cdot \mathbf{r}\right) = \chi\left(\hat{R}^{-1}\mathbf{r}\right). \tag{6}$$

This is a contravariant operator. To avoid potential obfuscation of symbolic calculus system, one can use Dirac notation instead,

$$(g \cdot \chi)(\mathbf{r}) = \langle\mathbf{r}|\, g \cdot |\chi\rangle = (\langle\mathbf{r}| \cdot g)\,|\chi\rangle = \left(g^{-1} \cdot |\mathbf{r}\rangle\right)^{\dagger} |\chi\rangle \tag{7}$$

and with $g \cdot |\mathbf{r}\rangle = |g \cdot \mathbf{r}\rangle$.

As for $V_{\nu\mathbf{q}}$, in order to apply representation theory, we need to interpret it as a element in some linear space, which requires to be clarify by tensor operator concept. Phonons are quasi-particle after second-quantization, so from the moment space perspective, there is no need to distinguish $G_{\mathbf{k}}$ and $G_{\mathbf{q}}$ in the mathematics. By linear approximation, we can say $V_{\nu\mathbf{q}}$ is a function or functional of $\mathbf{q}$, henceforth we deem $V_{\nu\mathbf{q}}$ as similar as $|\psi_{n\mathbf{k}}\rangle$ in representation theory level when calculating characters afterwards.

We have got all basic concepts to define our problems rigorously. The (1) is a element in $\mathrm{Hom}_H\left(V^*_{f\mathbf{k}+\mathbf{q}}\otimes V_{\nu\mathbf{q}}\otimes V_{i\mathbf{k}},\mathbb{C}\right)$ due to $|\psi_{i\mathbf{k}}\rangle = V_{i\mathbf{k}}$, $\hat{V}_{\nu\mathbf{q}}\in V_{\nu\mathbf{q}}$ ans so as $|\psi_{f\mathbf{k}}+\mathbf{q}\rangle$, here $V_{i\mathbf{k}}, V_{\nu\mathbf{q}}, V_{f\mathbf{k}+\mathbf{q}}$ are all irreducible with the assumption of no accidental degeneracy, and

$$H \coloneqq G_{\mathbf{k}+\mathbf{q}}\cap G_{\mathbf{q}}\cap G_{\mathbf{k}} \tag{8}$$

is the biggest stabilizer group of initial and final electron state and phonon state. By Schur's lemma, if $g^{\nu}_{fi}(\mathbf{k},\mathbf{q})\neq 0$, the group theory requires the Hom set is nonempty and then we have equivalence between two representation

$$V^*_{f\mathbf{k}+\mathbf{q}}\otimes V_{\nu\mathbf{q}}\otimes V_{i\mathbf{k}} \overset{H}{\sim} \mathbb{C}, \tag{9}$$

with the latter one being trivial representation. Finally, what we means when discussing symmetry permission is that we are asking is there trivial representation contained in $V^*_{f\mathbf{k}+\mathbf{q}}\otimes V_{\nu\mathbf{q}}\otimes V_{i\mathbf{k}}$? By character theory, the multiplicity with which trivial representation occurs in is

$$a_1 \coloneqq \dim_{\mathbb{C}}\mathrm{Hom}_H\left(V^*_{f\mathbf{k}+\mathbf{q}}\otimes V_{\nu\mathbf{q}}\otimes V_{i\mathbf{k}},\mathbb{C}\right), \tag{10}$$

one can solve this by calculating

$$a_1 = \frac{1}{|H|}\sum_{g\in H}\overline{\chi_{f\mathbf{k}+\mathbf{q}}(h)}\chi_{\nu\mathbf{q}}(h)\,\chi_{i\mathbf{k}}(h) = \begin{cases} 0, & \text{symmetry forbidden,}\\ \geq 1, & \text{symmetry allow.}\end{cases} \tag{11}$$

Here $\chi_{i\mathbf{k}}(h)$ is the character of i-th irreducible of canonical representation space $\mathcal{H}_{\mathbf{k}}$, so as $\chi_{f\mathbf{k}+\mathbf{q}}$ and $\chi_{\nu\mathbf{q}}$.

However, generally speaking, the group H is not finite as character theory fails to be applied here. There are 2 ways to tackle this problem:

(1) By projective representation theory. One can use group extension method to study SG,

$$0\to L\xrightarrow{\iota} SG\xrightarrow{\pi} PG\to 1, \tag{12}$$

with $L\lhd SG$ be the Abel extension kernel and point group $PG = SG/L$ be the quotient group. We have know that any representations of L correspondens to one counterpart in Pontryain dual and PG is finite group by Bieberbach's theorem and belongs to one of the widely known 32 point groups, which can be studied according to character theory. By synthesizing the representation of L and PG with group cohomology of, one can get those irreducible representations of SG in demand. By some guidance of 2-cohomology group $H^2(PG,L)$, we can find all the unequivalent way to extend a SG by given L and PG, thus irreducible representations.

(2) The Herring method. By observing the physical meaning of representation, we notice that the unnecessary redundance arise from translational symmetry. We can not just quotient out by L and study PG only because of potential nonsymmorphic symmetry generated by nontrivial extension, described by nontrivial element in

$H^2(PG, L)$ or nontrivial outer action of PG on L. The outer action is a homomorphism $c : PG \to \mathrm{Out}(L)$. But we can settle for the next best thing by quotient out by the biggest stabilizer group that leaves the 3 wave funtions $|\mathbf{k}\rangle$, $|\mathbf{q}\rangle$, $|\mathbf{k}+\mathbf{q}\rangle$ invariant.

3. The Herring Method

Definition 3.1. The *compatible sublattice* is defined as the least common multiple of period of $\chi_{\mathbf{k}}, \chi_{\mathbf{q}}, \chi_{\mathbf{k}+\mathbf{q}} \in \hat{L}$, i.e.

$$M := \ker\chi_{\mathbf{k}} \cap \ker\chi_{\mathbf{q}} \cap \ker\chi_{\mathbf{k}+\mathbf{q}}. \tag{13}$$

Let us look an example to grasp the physical meaning of M. Consider a 1 dimensional lattice system $L = \mathbb{Z}$, then $\forall \chi_k \in \widehat{L} = \mathrm{Hom}(L, U(1))$ is $\chi_k(n) = \mathrm{e}^{-\mathrm{i}2\pi kn}$, and thus $M = \ker\chi_k = \frac{1}{k}\mathbb{Z} \cap L$ and $[L : M] < \infty$. When it comes to irrational k, M can be a single point set $\{0\}$ and $[L : M] = \infty$.

Proposition 3.2. $[L : M] < \infty$ *if* $\mathbf{k}$, $\mathbf{q}$ *and* $\mathbf{k}+\mathbf{q}$ *are all in rational point in reciprocal space.*

Proof. By promoting the example of 1 dimensional to 3 dimensional, one can intuitively accomplish proof. □

Definition 3.3. *The Herring group* is defined as $F := H/M$.

The Herring group can be interpret as H with quotienting out of those redundant translational symmetry operations. Intuitively this method will not lose any necessary symmetry, both symmorphic and nonsymmorphic. And when we are focusing only on those rational momentum point, the Herring group is finite and hence be appropriate for applying characters.

The final theorem is to prove:

Theorem 3.4. *We have*

$$\mathrm{Hom}_H\left(V^*_{f\mathbf{k}+\mathbf{q}} \otimes V_{\nu\mathbf{q}} \otimes V_{i\mathbf{k}}, \mathbb{C}\right) = \mathrm{Hom}_F\left(V^*_{f\mathbf{k}+\mathbf{q}} \otimes V_{\nu\mathbf{q}} \otimes V_{i\mathbf{k}}, \mathbb{C}\right). \tag{14}$$

where $F = H/M$.

Proof. We need to show that for $\sigma : V^*_{f\mathbf{k}+\mathbf{q}} \otimes V_{\nu\mathbf{q}} \otimes V_{i\mathbf{k}} \to \mathbb{C}$,

$$(\forall h \in H : \sigma\phi_V(h) = \sigma) \iff \left(\forall \bar{h} \in F : \sigma\bar{\phi}_V\left(\bar{h}\right) = \sigma\right), \tag{15}$$

We can obtain this by showing that M acts trivially on $V^*_{f\mathbf{k}+\mathbf{q}} \otimes V_{\nu\mathbf{q}} \otimes V_{i\mathbf{k}}$. By definition 3.1 □

According to 14, we can use the latter to calculate a_1 by taking dimension

$$a_1 = \dim_{\mathbb{C}} \mathrm{Hom}_F\left(V^*_{f\mathbf{k}+\mathbf{q}} \otimes V_{\nu\mathbf{q}} \otimes V_{i\mathbf{k}}, \mathbb{C}\right) \tag{16}$$

$$= \frac{1}{|F|}\sum_{\bar{h}\in F} \overline{\chi^F_{f\mathbf{k}+\mathbf{q}}\left(\bar{h}\right)}\chi^F_{\nu\mathbf{q}}\left(\bar{h}\right)\chi^F_{i\mathbf{k}}\left(\bar{h}\right). \tag{17}$$

That is the core expression we need to calculate in scheme.

However, that is not the end. If we want to calculate the representation of F on $V_{f\mathbf{k}+\mathbf{q}}$, $V_{\nu\mathbf{q}}$, $V_{i\mathbf{k}}$ directly, we need to know the wave functions before and that can be only acquired by solving Kohn–Sham euqation in Density Functional Theory. We do not what to solve differential equation but with only space group information to determin the degeneracy.

To clarify the problem, after obtaining all the irreducible representations of F, we are unable to distinguish which ones are corresponding to given momentum label $\mathbf{k}$. When doing 17, we need to know the relation between momenta and irreducible representations.

Definition 3.5. $N \coloneqq L/M$.

Continuing with the previous 1 dimensional example, $N = L/\left(\frac{1}{k}\mathbb{Z} \cap L\right) = \mathbb{Z}/\left(\frac{1}{k}\mathbb{Z} \cap \mathbb{Z}\right)$. If $k = 1, 2, 3, \cdots$, then $N = 0$ is a trivial group; if $k = \frac{p}{q} \in \mathbb{Q}$ is a reduced fraction, then $M = q\mathbb{Z} \cong \mathbb{Z}$ and $N = \mathbb{Z}_q$. By the rational fraction case, we can interpret M as the smallest lattice or repetitive unit that contains the complete waveform and N as inequivalent lattice point with respective to waveform in on repetitive unit. And F is the "space group" of M, the reduced cell.

$N \trianglelefteq F$ because $L \trianglelefteq H$ then we can quotient both side of M. Let ϕ be a representation of F and $\phi|_N$ be the constraint upon N. By the completeness of the characters of N, we have

$$\phi|_N = \bigoplus_{\chi_{\mathbf{p}} \in \widehat{N}} m_{\mathbf{p}} \chi_{\mathbf{p}}, \tag{18}$$

where $\widehat{N}$ is the Pontryagin dual of N, $\chi_{\mathbf{p}}$ is the character labled by $\mathbf{p}$ and $m_{\mathbf{p}}$ is the multiplicity of $\chi_{\mathbf{p}}$. One can interpret $\mathbf{p}$ as the reduced momentum in reduced lattice N.

Theorem 3.6. *(Clifford, 1937) Let $\phi \in \operatorname{Irr}(F)$, then there is an common multiplicity e satisfies*

$$\phi|_N = e \bigoplus_{\mathbf{p}} \chi_{\mathbf{p}}, \tag{19}$$

and all the $\mathbf{p}$ belongs to one orbit under the group action of F.

Proof. Let $\phi_{\mathbf{p}} \in \widehat{N}$ be one component of $\phi|$ and V_ϕ be the representation space of ϕ, define

$$W_{\phi_{\mathbf{p}}} \coloneqq \{v \in V_\phi : \phi(n) v = \phi_{\mathbf{p}}(n) v, \forall n \in N\}. \tag{20}$$

Then $\forall h \in H$,

$$\begin{aligned} \phi(n)\phi(h) v &= \phi(h)\phi\left(h^{-1}nh\right) v = \phi_{\mathbf{p}}\left(h^{-1}nh\right)\phi(g) v \\ &= (h \cdot \phi_{\mathbf{p}})(n)\phi(g) v = [(h \cdot \phi_{\mathbf{p}})(n)]\, g \cdot v, \end{aligned} \tag{21}$$

which means

$$h \cdot W_{\phi_{\mathbf{p}}} = W_{h \cdot \phi_{\mathbf{p}}} = W_{\phi_{h \cdot \mathbf{p}}}. \tag{22}$$

Thus $\sum_{h\in H} h\cdot W_{\phi_{\mathbf{p}}}$ forms a F-invariant subspace of V_ϕ. By the irreducibility of V_ϕ,

$$\sum_{h\in H} h\cdot W_{\phi_{\mathbf{p}}} = V_\phi. \tag{23}$$

The h transitively permutate these $W_{\phi_{\mathbf{p}}}$'s, as a result, all of them have the same dimension, denote as e. □

Corollary 3.7. *Every $\phi\in \mathrm{Irr}\,(F)$ lands in one orbit generated by F-actions,*

$$\mathrm{Irr}\,(F) = \coprod_{\mathrm{Orb}\in\widehat{N}/F} \mathrm{Irr}_{\mathrm{Orb}}\,(F). \tag{24}$$

In our case, by definition (8), $F = H/M$ will leave any $\chi_{\mathbf{p}}$ invariant. Hence the orbit of $\chi_{\mathbf{p}}$ is just itself singly.

Definition 3.8. *The allowed representation* is defined as

$$\mathrm{Irr}_{\chi_{\mathbf{p}}}\,(F) \coloneqq \left\{\phi\in\mathrm{Irr}\,(F) : \phi\,(n) = \chi_{\mathbf{p}}\,(n)\cdot\mathrm{id}, \forall n\in N\right\}, \tag{25}$$

which means the $\phi\chi_F \in \mathrm{Irr}_{\chi_{\mathbf{p}}}\,(F)$ makes $\phi(h)$ a scalar matrix, $\forall h\in F$.

And that is exactly what we want — the $\mathrm{Irr}_{\chi_{\mathbf{p}}}\,(F)$ contains the irreducible characters of F behavior in correspondence with momentum $\mathbf{p}$ when it comes to translational operator in N.

We have found the irreducible representation of F can be classify according to momentum label $\mathbf{p}$, then how can we do this by using character table?

Lemma 3.9. *(trace saturation) Let A be a square Hermitian matrix with size d, then $|\mathrm{tr}\,A|\le d$, equality holds iff $A=\lambda\,\mathrm{id}$ with $\lambda\in U(1)$.*

Proof. The details are omitted; see the discussion in linear algebra textbooks. □

According to 3.9, one can find the $\mathbf{p}$ label of $\phi\in\mathrm{Iff}\,(F)$ by calculating

$$\frac{\chi_\phi\,(\bar{t}_j)}{\dim_{\mathbb{C}}\phi} = \mathrm{e}^{-2\pi\mathrm{i}p_j}, \qquad j=1,2,3, \tag{26}$$

where $\bar{t}_j$'s are 3 generators of N, and $\mathbf{p} = \begin{pmatrix} p_1 & p_2 & p_3\end{pmatrix}^{\mathrm{T}}$.

References